Introduction

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Abstract

This project develops a multiple linear regression model to estimate used vehicle selling prices from vehicle characteristics, condition, mileage, model, and geographic location. The resulting model explains approximately 82% of the variation in observed selling prices and provides a framework for supporting vehicle acquisition, pricing, and inventory decisions.

Keywords: Regression, Used Vehicles, Predictive Analytics, Vehicle Valuation

The objective of this project is to develop a predictive model capable of estimating the market value of used vehicles. Accurate vehicle valuation supports pricing decisions for dealerships, fleet operators, auction participants, and vehicle purchasing organizations.

The business decision addressed by this analysis is:

> Given a used vehicle's characteristics, what selling price should be expected in the current market?

The resulting model can be used to identify vehicles that appear underpriced or overpriced relative to market expectations.

1 Business Objective

The primary objective is to minimize pricing error between predicted and observed vehicle sale prices.

A practical loss function for this problem is Mean Squared Error (MSE):

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where:

- y_i is the observed selling price
- $\hat{y}_i$ is the predicted selling price

Lower prediction error improves decision quality by reducing the likelihood of overpaying for inventory or undervaluing vehicles during resale.

2 Plan

The analysis follows the regression modeling workflow:

1. Define the business problem.
2. Acquire and clean vehicle sales data.
3. Explore relationships between candidate predictors and selling price.
4. Build an interpretable regression model.
5. Evaluate assumptions and model fit.
6. Generate predictions and business recommendations.

The ideal dataset would contain:

- Vehicle age
- Odometer reading
- Mechanical condition
- Original MSRP
- Vehicle trim level
- Accident history
- Maintenance history
- Regional market conditions
- Vehicle Make/Model
- Sell price

Several of these variables were unavailable. Vehicle model and state were therefore used as proxy variables for differences in vehicle desirability and regional pricing effects. Vehicle make, age, and sell price should also act as a proxy for original MSRP.

The sales data was limited to 34 states. However, some states like Alaska had few observations.

3 Build

3.1 Data Source

The dataset was obtained from Kaggle at https://www.kaggle.com/code/bavlyy-oussef/vehicle-sales/input and contains used vehicle transaction information. It's unclear how this data was collected or where it came from exactly.

Variables initially available included:

- Year
- Body
- Color
- Make
- Model
- MMR
- VIN
- Seller
- Transmission
- State
- Condition
- Odometer
- Selling price

3.2 Data Cleaning

The following preprocessing steps were performed:

- Removed records with missing values in critical fields.
- Calculated vehicle age from model year.
- Converted odometer values into units of 10,000 miles.
- Standardized categorical values.
- Grouped low-frequency vehicle models into manufacturer-specific "Other" categories.
- Grouped low-frequency states into broader state categories.
- Created dummy variables for categorical predictors.

3.3 Sampling Strategy

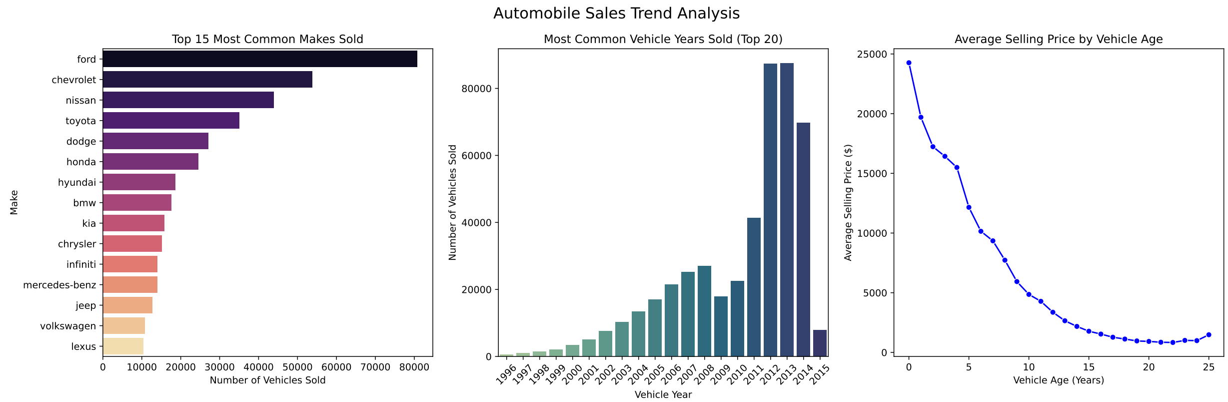
A stratified sampling approach was used to preserve state representation while reducing dataset size to 5000 for modeling performance.

Training and testing partitions were generated after sampling to maintain representative geographic distributions. A standard random sample was used to split off 20% of the data for testing.

4 Explore

4.1 Exploratory Data Analysis

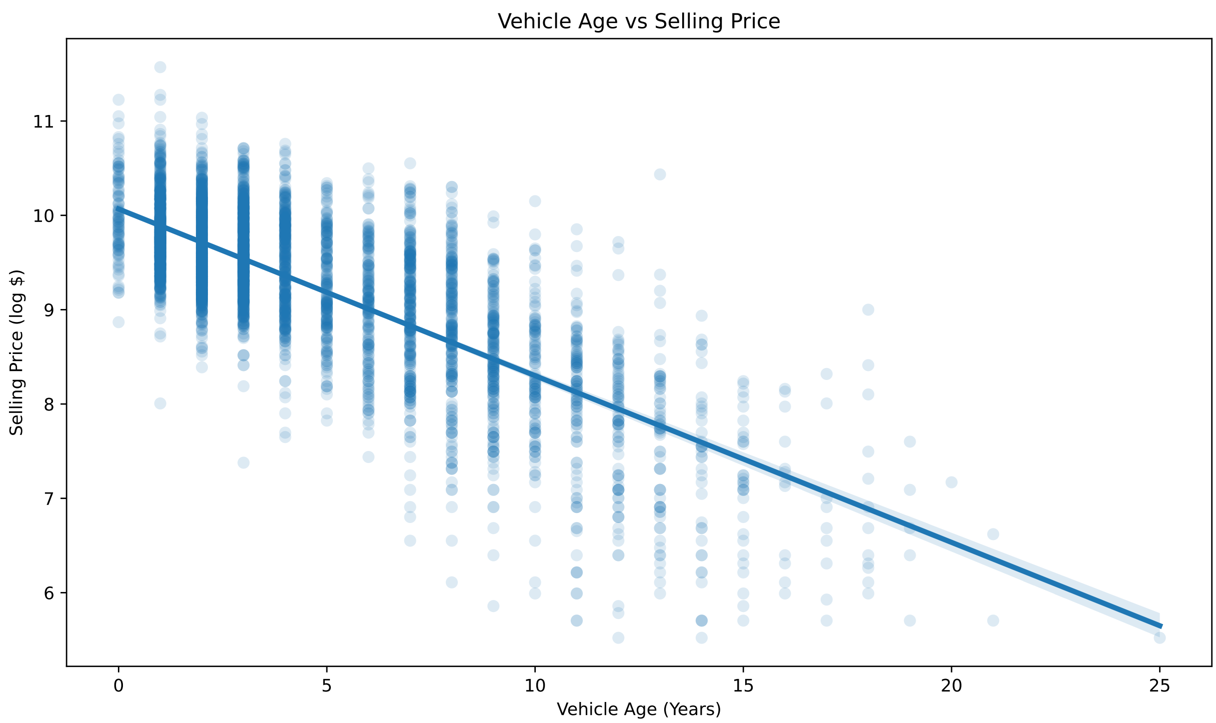
Initial visualizations focused on the relationships between selling price and major numerical predictors.



- A log transformation was applied to the selling price.

4.2 Vehicle Age vs Selling Price

Vehicle age showed a strong negative relationship with price. Older vehicles generally sold for substantially less than newer vehicles.



4.3 Odometer vs Selling Price

Vehicles with higher mileage generally sold for lower prices. The relationship was weaker than vehicle age but remained statistically significant.

4.4 Key Findings

- Vehicle age exhibits the strongest continuous predictor effect.
- Mileage has a meaningful negative effect on vehicle value.
- Vehicle model accounts for substantial differences in selling price.
- Trucks and heavy-duty pickups tend to command premium prices.
- Geographic location contributes measurable price variation.

5 Reconcile

5.1 Assumptions

Multiple linear regression requires:

- Linear relationships
- Independent observations
- Constant variance
- Approximately normal residuals
- Limited multicollinearity

5.2 Model Refinements

Several iterations of the model were evaluated.

Adjustments included:

- Consolidating sparse model categories.
- Grouping low-frequency states.
- Transforming odometer readings to improve interpretability.
- Evaluating Variance Inflation Factors (VIFs).

These modifications reduced model complexity while preserving predictive performance.

	Variable	VIF
0	const	178.803946
102	state_grouped_fl	6.907212
100	state_grouped_ca	6.474079
63	model_chevrolet_Other	6.085027

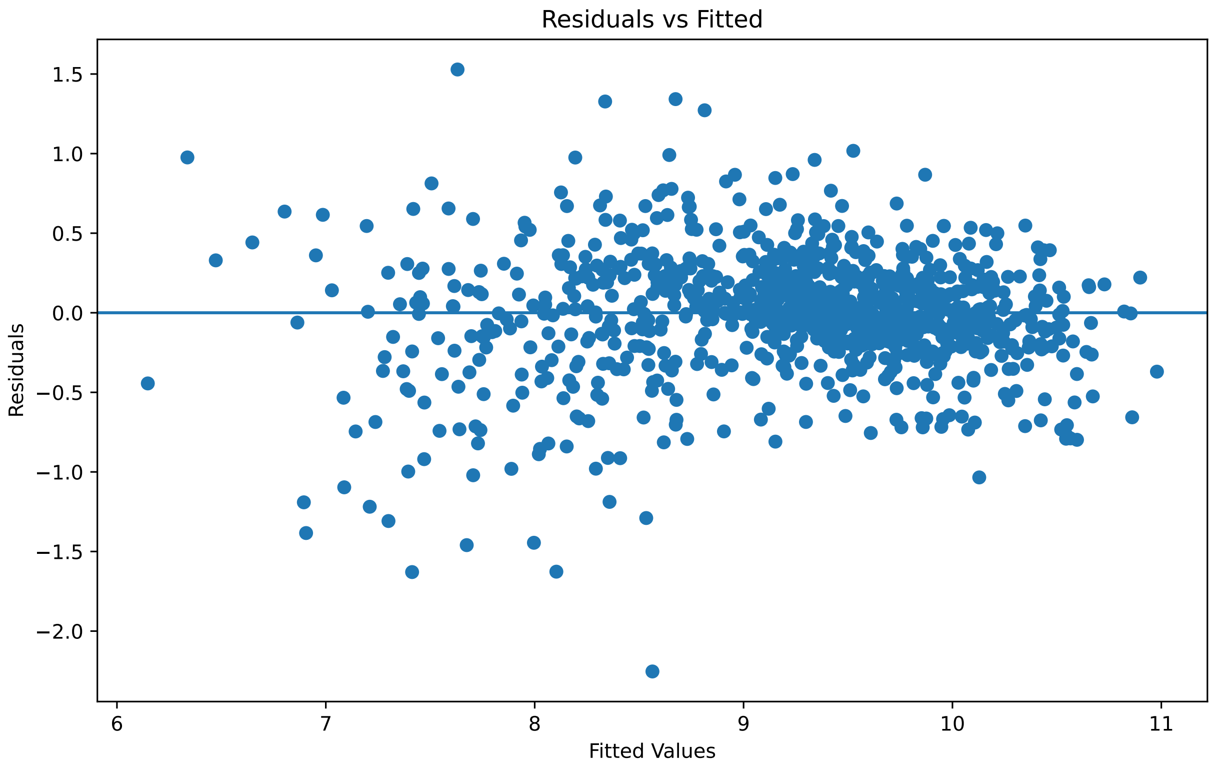
```
9           model_Altima    4.928018
..                   ...         ...
67        model_fiat_Other    1.193466
90        model_saab_Other    1.107573
93       model_smart_Other    1.103110
74       model_isuzu_Other    1.082286
65      model_daewoo_Other    1.040533

[124 rows x 2 columns]
```

Our original VIF showed high values with mmr, make, and body. Mmr is a market estimate for the worth of the vehicle so it made sense that predictor overpowered all of the other predictors. Likewise, it's a pretty safe assumption that the vehicle model was already capturing the make and body. For example a vehicle model of 'Ram 1500' only could refer to a dodge truck so the model can easily deduce that data from the vehicle model. Therefore including those in the model was redundant.

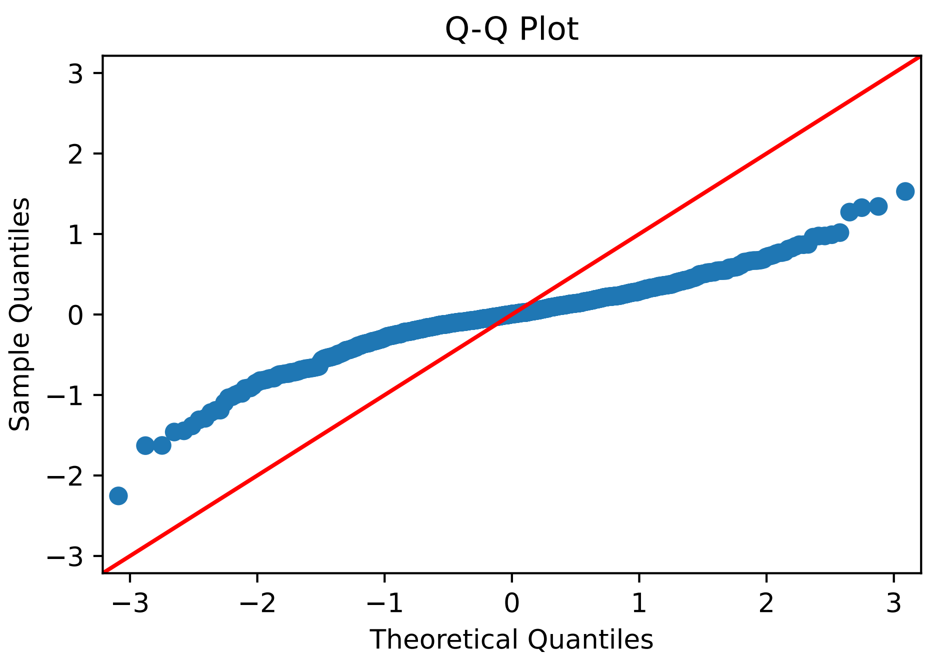
Our current model shows no VIF values above or close to 10 so there appear to be no concerns there.

```
Text(0.5, 1.0, 'Residuals vs Fitted')
```



Residuals remained approximately centered around zero with some evidence of heavy tails among high-value vehicles.

```
Text(0.5, 1.0, 'Q-Q Plot')
```



The Q-Q plot indicates that the residuals are not perfectly normally distributed. Rather than following the reference line, the residuals exhibit a noticeable S-shaped pattern, particularly in the tails. This suggests departures from the normality assumption and indicates that the distribution of errors is more concentrated around the mean than would be expected under a normal distribution.

Some departure from normality is expected in used vehicle sales data due to heterogeneous vehicle types, regional market differences, and the presence of categorical predictors such as vehicle model and state. While the log transformation of selling price improved the residual distribution substantially, the normality assumption is not fully satisfied.

Despite these deviations, the large sample size (n = 4,000) and the primary objective of prediction rather than inference make the model reasonably robust to moderate violations of residual normality. The residuals remain centered around zero, and other diagnostics did not indicate severe model deficiencies.

6 Fit

6.1 Final Model

The final model included:

- Vehicle age
- Odometer (10,000-mile units)
- Condition score
- Vehicle model
- State group

```
                            OLS Regression Results

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Dep. Variable:           sellingprice   R-squared:                       0.818

Model:                            OLS   Adj. R-squared:                  0.812

Method:                 Least Squares   F-statistic:                     141.6

Date:                Sat, 20 Jun 2026   Prob (F-statistic):               0.00

Time:                        20:42:47   Log-Likelihood:                -

1715.4

No. Observations:                4000   AIC:                             3679.

Df Residuals:                    3876   BIC:                             4459.

Df Model:                         123

Covariance Type:            nonrobust

==================================================================================

                             coef    std err          t      P>|t|      [0.025

----------------------------------------------------------------------------------

const                     10.0407      0.080    125.821      0.000       9.884

vehicle_age               -0.1279      0.003    -44.877      0.000      -

0.134      -0.122

odometer_10k              -0.0508      0.002    -24.010      0.000      -

0.055      -0.047

condition                  0.0073      0.001     14.577      0.000       0.006

model_200                 -0.6800      0.097     -6.995      0.000      -

0.871      -0.489

model_3 Series             0.0911      0.082      1.113      0.266      -
```

0.069 0.252					
model_300	-0.0872	0.100	-0.876	0.381	-
0.282 0.108					
model_5 Series	0.3066	0.099	3.104	0.002	0.113
model_Accord	-0.2591	0.082	-3.175	0.002	-
0.419 -0.099					
model_Altima	-0.4687	0.075	-6.227	0.000	-
0.616 -0.321					
model_Avenger	-0.7208	0.091	-7.921	0.000	-
0.899 -0.542					
model_C-Class	0.0999	0.090	1.116	0.264	-
0.076 0.275					
model_CR-V	0.1006	0.107	0.941	0.347	-
0.109 0.310					
model_Camry	-0.3019	0.078	-3.865	0.000	-
0.455 -0.149					
model_Charger	-0.0663	0.098	-0.673	0.501	-
0.259 0.127					
model_Civic	-0.4153	0.086	-4.853	0.000	-
0.583 -0.248					
model_Corolla	-0.4822	0.084	-5.756	0.000	-
0.646 -0.318					
model_Cruze	-0.6591	0.090	-7.338	0.000	-
0.835 -0.483					

model_E-Class 0.3532 0.096 3.673 0.000 0.165

model_Edge -0.0024 0.090 -0.027 0.978 -

0.179 0.174

model_Elantra -0.6615 0.086 -7.688 0.000 -

0.830 -0.493

model_Equinox -0.2747 0.087 -3.153 0.002 -

0.446 -0.104

model_Escape -0.2795 0.077 -3.647 0.000 -

0.430 -0.129

model_Expedition 0.1691 0.109 1.558 0.119 -

0.044 0.382

model_Explorer 0.0166 0.085 0.195 0.845 -

0.150 0.184

model_F-150 0.2495 0.078 3.181 0.001 0.096

model_F-250 Super Duty 0.5993 0.100 5.968 0.000 0.402

model_Focus -0.7664 0.078 -9.873 0.000 -

0.919 -0.614

model_Fusion -0.4933 0.077 -6.412 0.000 -

0.644 -0.342

model_G Sedan -0.0277 0.084 -0.331 0.740 -

0.192 0.136

model_Grand Caravan -0.3861 0.086 -4.490 0.000 -

0.555 -0.218

model_Grand Cherokee 0.0804 0.097 0.832 0.406 -

```
0.109       0.270

model_Impala                 -0.5320      0.084      -6.326      0.000      -

0.697      -0.367

model_Jetta                  -0.5260      0.100      -5.246      0.000      -

0.723      -0.329

model_Journey                -0.3347      0.095      -3.531      0.000      -

0.521      -0.149

model_Liberty                -0.1972      0.109      -1.817      0.069      -

0.410       0.016

model_Malibu                 -0.5346      0.085      -6.321      0.000      -

0.700      -0.369

model_Maxima                 -0.2231      0.087      -2.569      0.010      -

0.393      -0.053

model_Mazda3                 -0.5291      0.099      -5.332      0.000      -

0.724      -0.335

model_Mustang                -0.2246      0.103      -2.187      0.029      -

0.426      -0.023

model_Odyssey                -0.0549      0.098      -0.562      0.574      -

0.247       0.137

model_Optima                 -0.4698      0.094      -4.988      0.000      -

0.654      -0.285

model_Passat                 -0.6219      0.094      -6.632      0.000      -

0.806      -0.438

model_RAV4                   -0.0871      0.104      -0.836      0.403      -
```

0.291 0.117

model_Rogue -0.3026 0.096 -3.151 0.002 -

0.491 -0.114

model_Sentra -0.6501 0.090 -7.223 0.000 -

0.827 -0.474

model_Sienna 0.1062 0.107 0.991 0.322 -

0.104 0.316

model_Silverado 1500 0.2541 0.087 2.932 0.003 0.084

model_Sonata -0.5277 0.088 -6.028 0.000 -

0.699 -0.356

model_Sorento -0.3940 0.103 -3.843 0.000 -

0.595 -0.193

model_Soul -0.6376 0.105 -6.050 0.000 -

0.844 -0.431

model_Suburban 0.4482 0.104 4.309 0.000 0.244

model_Tahoe 0.5444 0.100 5.462 0.000 0.349

model_Taurus -0.6687 0.105 -6.360 0.000 -

0.875 -0.463

model_Town and Country -0.3667 0.093 -3.946 0.000 -

0.549 -0.184

model_Versa -0.7535 0.108 -6.969 0.000 -

0.965 -0.542

model_Wrangler 0.4013 0.092 4.343 0.000 0.220

model_X5 0.4439 0.108 4.094 0.000 0.231

model_acura_Other	0.1363	0.089	1.535	0.125	-0.038	0.310
model_audi_Other	0.2036	0.088	2.318	0.021	0.031	
model_bmw_Other	0.3983	0.087	4.569	0.000	0.227	
model_buick_Other	-0.3924	0.093	-4.213	0.000	-0.575	-0.210
model_cadillac_Other	-0.0152	0.081	-0.186	0.852	-0.175	0.145
model_chevrolet_Other	-0.2908	0.073	-3.962	0.000	-0.435	-0.147
model_chrysler_Other	-1.1530	0.099	-11.659	0.000	-1.347	-0.959
model_daewoo_Other	-2.2839	0.385	-5.931	0.000	-3.039	-1.529
model_dodge_Other	-0.1590	0.080	-1.997	0.046	-0.315	-0.003
model_fiat_Other	-0.9297	0.168	-5.519	0.000	-1.260	-0.599
model_ford_Other	-0.2788	0.078	-3.562	0.000	-0.432	-0.125
model_gmc_Other	0.2408	0.083	2.918	0.004	0.079	
model_honda_Other	-0.0278	0.097	-0.287	0.774	-0.217	0.162
model_hummer_Other	0.8827	0.144	6.140	0.000	0.601	

model_hyundai_Other	-0.4713	0.084	-5.607	0.000	-
0.636	-0.306				
model_infiniti_Other	0.2024	0.084	2.407	0.016	0.038
model_isuzu_Other	-0.9221	0.278	-3.320	0.001	-
1.467	-0.378				
model_jaguar_Other	0.2494	0.129	1.935	0.053	-
0.003	0.502				
model_jeep_Other	-0.3907	0.090	-4.343	0.000	-
0.567	-0.214				
model_kia_Other	-0.7087	0.083	-8.545	0.000	-
0.871	-0.546				
model_land rover_Other	0.6535	0.129	5.075	0.000	0.401
model_lexus_Other	0.3985	0.080	4.990	0.000	0.242
model_lincoln_Other	-0.1519	0.093	-1.631	0.103	-
0.334	0.031				
model_mazda_Other	-0.5526	0.089	-6.230	0.000	-
0.727	-0.379				
model_mercedes-benz_Other	0.5828	0.088	6.641	0.000	0.411
model_mercury_Other	-0.4185	0.138	-3.042	0.002	-
0.688	-0.149				
model_mini_Other	-0.3755	0.107	-3.515	0.000	-
0.585	-0.166				
model_mitsubishi_Other	-0.6594	0.094	-6.988	0.000	-
0.844	-0.474				

model_nissan_Other	-0.1042	0.078	-1.333	0.183	-
0.258	0.049				
model_pontiac_Other	-0.7952	0.097	-8.183	0.000	-
0.986	-0.605				
model_porsche_Other	1.0412	0.150	6.931	0.000	0.747
model_ram_Other	0.3523	0.158	2.224	0.026	0.042
model_saab_Other	-0.7784	0.229	-3.393	0.001	-
1.228	-0.329				
model_saturn_Other	-0.7523	0.101	-7.435	0.000	-
0.951	-0.554				
model_scion_Other	-0.5546	0.150	-3.701	0.000	-
0.848	-0.261				
model_smart_Other	-1.1332	0.229	-4.949	0.000	-
1.582	-0.684				
model_subaru_Other	-0.1688	0.096	-1.750	0.080	-
0.358	0.020				
model_suzuki_Other	-0.6913	0.138	-5.004	0.000	-
0.962	-0.420				
model_toyota_Other	0.1528	0.077	1.994	0.046	0.003
model_volkswagen_Other	-0.2980	0.093	-3.221	0.001	-
0.479	-0.117				
model_volvo_Other	-0.2249	0.098	-2.284	0.022	-
0.418	-0.032				
state_grouped_az	0.0218	0.062	0.349	0.727	-

0.100	0.144				
state_grouped_ca	0.1581	0.044	3.599	0.000	0.072
state_grouped_co	0.0942	0.067	1.399	0.162	-
0.038	0.226				
state_grouped_fl	0.1232	0.043	2.842	0.005	0.038
state_grouped_ga	0.1410	0.047	3.004	0.003	0.049
state_grouped_il	0.1443	0.050	2.909	0.004	0.047
state_grouped_in	0.0447	0.078	0.574	0.566	-
0.108	0.197				
state_grouped_ma	-0.0426	0.066	-0.648	0.517	-
0.171	0.086				
state_grouped_md	-0.0296	0.059	-0.503	0.615	-
0.145	0.086				
state_grouped_mi	0.1618	0.054	3.024	0.003	0.057
state_grouped_mn	0.2002	0.060	3.332	0.001	0.082
state_grouped_mo	0.1418	0.053	2.678	0.007	0.038
state_grouped_nc	0.1157	0.051	2.268	0.023	0.016
state_grouped_ne	0.1334	0.080	1.676	0.094	-
0.023	0.290				
state_grouped_nj	0.0773	0.049	1.565	0.118	-
0.020	0.174				
state_grouped_nv	0.1024	0.056	1.836	0.066	-
0.007	0.212				
state_grouped_ny	0.0479	0.072	0.661	0.508	-

```
0.094      0.190

state_grouped_oh           0.0613     0.050     1.225     0.221     -

0.037      0.159

state_grouped_pa           0.1496     0.049     3.054     0.002      0.054

state_grouped_sc           0.1564     0.076     2.059     0.040      0.007

state_grouped_tn           0.1271     0.050     2.528     0.011      0.029

state_grouped_tx           0.0745     0.045     1.646     0.100     -

0.014      0.163

state_grouped_va           0.0361     0.059     0.617     0.537     -

0.079      0.151

state_grouped_wa           0.0933     0.064     1.462     0.144     -

0.032      0.218

state_grouped_wi           0.1179     0.059     1.989     0.047      0.002

==============================================================================

Omnibus:                      499.223   Durbin-Watson:                   2.020

Prob(Omnibus):                  0.000   Jarque-Bera (JB):             3013.924

Skew:                          -0.431   Prob(JB):                         0.00

Kurtosis:                       7.164   Cond. No.                     3.85e+03

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Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specifi

[2] The condition number is large, 3.85e+03. This might indicate that there are

strong multicollinearity or other numerical problems.
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The fitted model produced:

- $R^2 = 0.818$
- Adjusted $R^2 = 0.812$
- F-statistic = 141.6
- $p < 0.001$

indicating strong overall explanatory power.

6.2 Important Continuous Predictors

Variable	Coefficient
Vehicle Age	-0.1279
Odometer (10k miles)	-0.0508
Condition	0.0073

Because the dependent variable is log-transformed, coefficients can be approximately interpreted as percentage changes.

6.2.1 Vehicle Age

Each additional year reduces expected selling price by approximately:

$$100(e^{-0.1279} - 1) \approx -12.0\%$$

6.2.2 Odometer

Each additional 10,000 miles reduces expected selling price by approximately:

$$100(e^{-0.0508} - 1) \approx -5.0\%$$

6.2.3 Condition

Each one-point increase in condition increases expected selling price by approximately:

$$100(e^{0.0073} - 1) \approx 0.7\%$$

7 Evaluate

7.1 Residual Analysis

Residual diagnostics indicated:

- No severe violations of linearity.
- Acceptable residual symmetry.
- Some heavy tails remain.
- No extreme multicollinearity concerns.

The largest VIF values were associated with popular truck categories but remained below commonly cited thresholds requiring corrective action.

7.2 Limitations

Several limitations should be noted:

- No MSRP information was available.
- Trim level information was not incorporated.

- Accident history was unavailable.
- Maintenance records were unavailable.
- Auction data may not represent all vehicle markets.

These limitations likely explain a portion of the unexplained variance.

8 Predict

8.1 Business Implications

The model suggests:

1. Vehicle age is the strongest continuous driver of depreciation.
2. Mileage has a significant but smaller effect.
3. Vehicle model contributes substantial value differences.
4. Trucks generally retain value better than passenger sedans.

8.2 Recommendations

Organizations purchasing used vehicles should:

- Prioritize lower-age vehicles when possible.
- Consider mileage alongside age rather than in isolation.
- Use model-specific adjustments when evaluating value.
- Compare observed asking prices against model predictions to identify unusually favorable purchasing opportunities.

8.3 Future Work

Future improvements could include:

1. Incorporating original MSRP.
2. Including trim-level information.
3. Adding accident-history variables.
4. Incorporating local economic indicators.

9 Conclusion

The final regression model explains approximately 82% of the variation in used vehicle selling prices and provides a practical framework for vehicle valuation. Vehicle age, mileage, model, and geographic location all contribute significantly to price determination. The resulting model can support data-driven purchasing and pricing decisions while highlighting areas for future model enhancement.

References